

Group Project Instructions

1. Purpose

You will produce an original, evidence-based study of one part of the business analytics (BA) career landscape as it stands in 2026. You collect the data, analyze it, and deliver a recommendation to a named audience. Two purposes. First, **integration**: in MGS 2150, 3001, 3101, 3501 and 3701 you applied techniques one at a time to data someone else had cleaned, for questions someone else had framed. Here you frame the question, obtain the data, choose the methods, and defend the choice. Second, **preparation**: every track answers something you need to know before you graduate — what employers want, what interviews test, what master's programs teach, and what admissions committees ask.

2. Data is the point

Ask a chatbot how generative AI has changed the BA job market and you get a fluent answer in four seconds. Everyone has that answer, so it is worth nothing. Evidence is the scarce thing, and it is what you are graded on. Every substantive claim in your report must trace to a specific row in a dataset you assembled or a document you can produce. A claim you cannot trace is a claim you delete. Reports, news and published research frame your question and cross-check your numbers; they do not replace them. Use only the methods your question needs — an honest descriptive comparison beats a causal claim your design cannot support.

3. Start collecting now

Collection will be harder than you expect. LinkedIn, BOSS 直聘, 小红书 and similar platforms use rate limiting, IP blocking, slider verification and login walls, and change page structures without notice. The usual pattern is three weeks lost to discovering this, then rushed analysis in the final fortnight.

- Attempt your primary source before Lab 1 on 15 September — a real script, a real result or error message. Bring it to the lab.
- Register a fallback in A1. Switching to it carries no penalty. Not having one does.
- A1 requires **at least 100 records** already collected, with the code or coding sheet that produced them.

A blocked source is itself a finding worth a paragraph: how a market gates its data tells you something about that market.

4. Teams and tracks

Five teams of four to five. One track per team; no duplicates. Below, each subsection provides the

topic, the audience, what you must establish, and a **suggested** data ladder: what to try first, what to do when it fails, last option. If you find a better source, use it and tell me.

4.1 Track 1 — The BA labor market in China

- **Example question.** What BA roles are being hired for in China in 2026, what do they require, what do they pay, and what has changed since generative AI?
- **Audience.** A WKU BA junior choosing what to learn and which roles to target for 2027 内推 or 校招.
- **Establish.** A role taxonomy from observed job titles; the skill mix per role family and how often AI-related skills appear; variation by city tier, firm type and industry; and the relationship between stated requirements and stated salary.
- **Data collection.**
 - 1) **Try:** BOSS 直聘 public search pages. Attempt 猎聘, 智联招聘, 前程无忧 and 实习僧 in parallel; their defenses differ. Nothing behind a login.
 - 2) **If blocked:** manual collection on a fixed schema (title, company, industry, city, salary range, experience, education, JD text, date, URL), 50–80 postings per member. Design the sampling frame before you collect.
 - 3) **Triangulate:** market reports from 智联招聘, 猎聘 and BOSS 直聘研究院; 国家统计局 for macro context. Checks on your data, not replacements.
- **Note.** Archived snapshots of Chinese job pages are patchy. If a clean before-and-after is unavailable, say so and use a cross-section instead — AI-adjacent versus traditional roles, or technology versus traditional firms.

4.2 Track 2 — The international BA labor market

- **Example question.** What BA roles are being hired for outside China in 2026, what do they require, what do they pay, and how has the skill mix shifted since 2023?
- **Audience.** A WKU BA senior weighing overseas study followed by an overseas or returnee job search.
- **Establish.** The same structure as Track 1, plus a genuine before-and-after: a large 2023–2024 posting dataset exists publicly and you can collect the 2026 equivalent.
- **Data collection.**
 - 1) **Try:** LinkedIn public job-search pages without login. Expect blocking; time-box to one week.
 - 2) **If blocked:** The Adzuna API — free key, roughly 1,000 calls per month, 16+ countries, with search, salary histogram, salary history and top-company endpoints. Descriptions are truncated and some salaries are modelled rather than stated (salary_is_predicted); treat those as a separate category.
 - 3) **Also try:** company job boards hosted on Greenhouse, Lever or Workday. Scraping fifty target employers directly yields cleaner data.

- 4) **Floor:** Kaggle “LinkedIn Job Postings (2023–2024)” by arshkon, with O*NET technology skills and BLS OEWS wage data. Check the license and cite it.
- **Note.** Your time comparison runs across two differently collected samples. Think about how to handle that comparability problem.

4.3 Track 3 — *What actually happens in a BA job interview*

- **Example question.** How are candidates for BA, DA and analytics internship roles assessed in 2026, and has AI-tool fluency entered the assessment?
- **Audience.** This cohort and the next.
- **Establish.** Round structure and what each round tests; the distribution of question types across SQL, statistics, business case, product sense, behavioral and AI-tool use; variation by firm type and role level; and any visible shift since 2023.
- **Data collection.**
 - 1) **Try:** 牛客网 面经 pages, much of which is readable without login — check robots.txt first. In parallel, the official Reddit API via PRAW (r/datascience, r/analytics).
 - 2) **If blocked:** 小红书 和 知乎 by manual collection into a structured coding sheet. You need hand-coded material for validation anyway.
 - 3) **Floor:** Reddit alone yields a sufficient corpus. This track cannot fail for want of data.
- **Note.** People who fail interviews post about them less often than people who pass. Your corpus is not a random sample and you must not describe it as one.

4.4 Track 4 — *The master’s programme landscape*

- **Example question.** How have BA and DS master’s programmes changed their curricula, prerequisites and admissions requirements since 2021, and which fit a WKU BA graduate?
- **Audience.** The WKU BA cohort applying between October and December — many of you.
- **Establish.** A dataset of about 60–80 programmes covering curriculum composition, prerequisites, test policy, tuition, duration and published outcomes; the shift in curriculum composition over five years; variation by region and by school type; and a defensible shortlist.
- **Data collection**
 - 1) **Try:** the programme websites directly. Universities rarely block scrapers. Build the sampling frame from published rankings.
 - 2) **For change over time:** The Wayback Machine and its CDX API. Snapshots of the same curriculum page from 2021 and 2026 are the whole comparison; there is no other route to it. If not available, comparison across time is not required.
 - 3) **Supplementary:** employment report PDFs for table extraction; 中国研究生招生信息网 for domestic programmes.

- **Note.** Published employment statistics are self-reported and inconsistently defined. Compared naively they rank reporting practices, not programmes.

4.5 Track 5 — *The graduate admissions interview*

- **Example question.** What do admissions interviews for BA and DS master's programmes assess, how does this differ by region and programme type, and how should a WKU applicant prepare?
- **Audience.** Yourself, Oct – Dec.
- **Establish.** Which programmes interview and in what format — live, panel or asynchronous video; the mix of motivation, technical, case and AI-ethics questions; variation by region and programme type; and a preparation protocol grounded in evidence rather than in what advice accounts assert.
- **Data collection**
 - 1) **Try:** programme admissions pages, many of which state the format and platform, plus Reddit r/gradadmissions via the official API.
 - 2) **If blocked:** 小红书 and 一亩三分地. Both are effectively login-walled; plan for manual collection from the start.
 - 3) **Alternative data (last option):** 8–12 semi-structured interviews with WKU alumni and current applicants.

5. Deliverables (A1, A2, A3, Final Presentation)

5.1 A1 — *Project brief and pilot data (10%)*

Due: Monday 28 September, 23:59.

Submit a project brief of **no more than 4 pages** (.doc), excluding the title page. The title page lists the project title, team members, your research question, and the link to your GitHub repository. The brief states what you intend to find out and how; the repository shows that data collection is already underway. Suggested contents of the brief:

- **Question and audience:** Your question, narrower than the track heading; who your answer is for and what decision it informs.
- **Sampling frame:** Population, search terms, time window, inclusion and exclusion rules, and the reasoning behind each.
- **Data collection:** Your primary source, the documented result of your first attempt at it, and your registered fallback if that source proves unworkable.
- **Risk:** One paragraph on what could make this project fail and how would your response
- **Division of labor:** A named owner for collection, cleaning, analysis and writing.
- **AI use log,** reference if any.

Required in the repository by the deadline:

- A pilot dataset of at least 100 records.
- The Jupyter notebooks that produced it. For manually collected data, the coding sheet and a written collection protocol.
- A README stating what the data are, where they came from, when they were collected, and how to reproduce the pilot.

A brief without pilot data and the code behind it in the GitHub repository is an incomplete submission.

5.2 A2 — *Analysis-ready dataset, validation and exploratory findings (10%)*

Due: Wednesday 4 November, 23:59.

Submit a findings report of no more than 10 pages (.doc), excluding the title page. The report shows what your data now show; the repository holds the dataset, the code/Jupyter notebooks and the evidence that your extraction is trustworthy. Suggested contents of the report:

- **Question:** Your question as it now stands. If it has changed since A1, say what changed and why.
- **Dataset:** Record count, coverage, collection period, and what is missing.
- **Validation:** Agreement between your automated extraction and the hand-coded sample for text analysis; inter-coder agreement between the two coders; where they disagree and what that implies for your findings.
- **Findings:** Several descriptive findings, each with appropriate figures.
- **Working answer:** What you currently believe the answer is, and the evidence that would show you were wrong.
- **AI use log,** reference if any.

Required in the repository by the deadline:

- Both the raw and cleaned datasets. If the files are too large to upload, convert them to a lighter format or split them into smaller files.
- A hand-coded gold set of at least 150 items, with the codebook used to code it.
- Collection, cleaning and extraction code in Jupyter notebooks, runnable end to end.
- Data documentation: source, collection dates, record count, field definitions, known gaps and known biases in the README file.
- A validation notebook that produces the agreement figures reported in the report.

A report showing validation figures that no notebook in the repository produces is an incomplete submission.

5.3 A3 — *Final report (20%)*

Due: Thursday 17 December, 23:59.

Submit the final report (.doc) and a repository release tagged final. Every claim in the report must be checkable against the repository. There is no template and no page limit. Each team designs the structure that fits its question and audience. Length is not a grading criterion and a longer report is not a better one; **~15 pages excluding appendices** is a reasonable target, with figures and tables counted inside it. Teams running substantially longer are usually padding a thin finding.

Whatever structure you choose, a reader must be able to find the following without hunting:

- A one-page executive summary, placed first, written for someone who does not know what a regression is.
- The question and the audience whose decision it informs.
- Data provenance: what you collected, from where, over what period, and what you excluded.
- Validation of your extraction against the hand-coded sample.
- Findings, each traceable to your data.
- Limitations and a recommendation with a success measure.
- Reference, if any.
- Appendices, which fall outside the suggested length: data documentation, AI use log, contribution statement, repository link.

Labs 3 and 4 apply causal inference and prescriptive analytics to your own data. Include what your question supports — a diagnostic decomposition, an identification strategy, a constrained decision problem. Where the data do not support a causal claim, say so and explain what would be needed. You are graded on the fit between question, method and evidence, not on how many techniques appear.

Required in the repository, tagged final:

- All codes and Jupyter notebooks, runnable from collection through to the figures in the report.
- Cleaned data where licensing permits; where it does not, the documentation and code that would regenerate it.
- A README summarizing the project in one paragraph and giving instructions for running the analysis.
- AI use log and individual contribution statement.

5.4 Final presentation and defense (20%)

10 December (Teams 1–3) and **15 December** (Teams 4–5). Draft slides due 7 December for Consultation 3 on 8 December.

Each team has a 25-minute slot: 20 minutes presenting and 5 minutes of questions. The presentation is a defense, not a report reading — assume the audience knows the track and wants

to know what you found and whether to believe it.

- Every member speaks, and each is graded individually on the defense as well as on the team's presentation.
- Questions may go to any member about any part of the work, including parts they did not write.
- Cover the question, how you obtained the data and what you did when the primary source failed, your validation, the findings, the limitations, and the recommendation.
- Slides brought to Consultation 3 need not be finished, but must be complete enough to show the argument's structure.

6. AI Policy

The course policy applies in full: disclosure, verification, attribution. You are expected to use AI tools; you are not permitted to let them do the thinking.

- **Encouraged:** writing and debugging code; proposing approaches for you to evaluate; translation; extraction and classification, provided the output is validated against your hand-coded sample; improving prose you have written.
- **Not permitted:** generating findings, conclusions or recommendations; producing, imputing or extending data; writing the judgement in your executive summary; submitting unvalidated extraction output.
- **Required:** an AI use log in your appendix — tool, task, what it produced, how you verified it. “Used ChatGPT for help with code” does not satisfy this.

The defense is where this is tested. If you cannot explain why your pipeline makes a decision, or what a coefficient in your own table means, authorship is in question whatever the log says.

7. How AI was used to write this document

This course asks you to disclose your use of AI, so here is mine. I drafted these instructions with Claude (Anthropic) across several sessions: structuring the tracks, drafting and redrafting prose, proposing candidate data sources, and checking the document against the syllabus for conflicts.

The decisions are mine — the purposes, the tracks and their scope, the deliverables and weights, grading criteria, the peer-evaluation rule, the conduct rules. Several drafts were rejected; one proposed track was removed entirely. Named data sources were verified rather than taken on trust, and proposals that sounded plausible but could not be confirmed were dropped. API limits, dataset licenses and platform terms change often enough that **you should confirm them yourself at the point of use.**

That pattern is the one this project asks of you. The model accelerated the drafting; it did not decide what the project should be, and nothing it asserted was kept without checking.

8. Calendar

Date	Milestone
Thu 3 Sep	Project introduced; team formation drafted
Tue 15 Sep	Lab 1 — scraper build. Arrive with a first attempt made
Mon 28 Sep	A1 due, 23:59
Tue 29 Sep	Consultation 1 — collection gate
Tue 20 Oct	Lab 2 — text analytics on the team corpus
Thu 29 Oct	GitHub repository initialized
Wed 4 Nov	A2 due, 23:59
Thu 5 Nov	Consultation 2 — dataset gate
Thu 19 Nov	Lab 3 — identification strategy on team data
Tue 1 Dec	Lab 4 — constrained decision problem
Mon 7 Dec	Draft slides
Tue 8 Dec	Consultation 3 — presentation gate
Thu 10 Dec	Final presentations, Teams 1–3
Tue 15 Dec	Final presentations, Teams 4–5
Thu 17 Dec	A3 due, 23:59